

Build a Baselet!Age range: 8-18 years

- Duration: 30–45 minutes

Purpose Statement

Students are at a key stage of scientific development, yet essential abstract concepts can feel purely theoretical and disconnected from real life. This hands-on activity uses a bracelet model to make transcription and translation tangible. By introducing these ideas in something the students can manipulate and take home, we want to help them develop a stronger understanding of how DNA encodes life. At the same time, we want to encourage curiosity about how small changes in genetic sequences can have meaningful biological consequences. This activity specifically focuses on the central dogma of biology, including how DNA stores genetic information, how RNA carries and translates that information, and how proteins are synthesized to perform essential functions within cells.

Why

Unlike worksheets or diagrams, this bracelet activity allows students to physically build and manipulate the genetic code. This hands-on learning will strengthen their conceptual understanding and teach them about the cause and effect relationships in genetics. Additionally, students leave with a wearable visual model that extends learning beyond the classroom.

Standards Alignment

This activity aligns with Next Generation Science Standards (NGSS). See full evidence statements at:
NGSS HS-LS1 Evidence Statements

- **HS-LS1-1**
  - 1a. Students construct an explanation that includes the idea that regions of DNA called genes determine the structure of proteins, which carry out the essential functions of life through systems of specialized cells.
  - 2a. Students identify and describe the evidence to construct their explanation, including that: i. All cells contain DNA; ii. DNA contains regions that are called genes; iii. The sequence of genes contains instructions that code for proteins

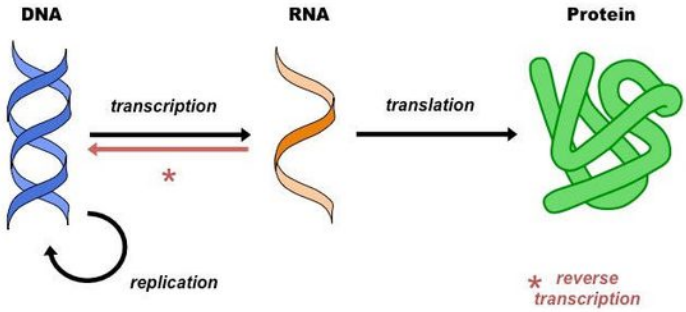
Background Information
This might be helpful to review before going through the activity with your student(s).

What is DNA?
DNA is found in all living cells, and contains all the information required for life. Just as we store information in letters spelling out words, DNA stores information using 4 nucleotide bases: Adenine(A), Thymine(T), Cytosine(C), and Guanine(G). These bases are then read and interpreted by other molecules to form living organisms

What is RNA?
RNA is the intermediate translator between DNA and proteins. In a process called transcription, enzymes read DNA's written code, and make a complementary RNA molecule. Like DNA, RNA codes information in 4 nucleotide bases, but uses Uracil(U) instead of Thymine. This RNA molecule then transports the information that DNA stores.

What are Proteins?
Proteins are the molecular machinery of the cell that perform a wide variety of functions from breaking down food, to moving muscles. Proteins are made up of a long strand of amino acids, the sequence of those amino acids determine the proteins structure and function. In a process called translation, the ribosome, or the protein factory of the cell, reads RNA to produce a specific protein encoded by it. It does this by reading 3 bases at a time which forms a codon. These codons each correspond to a specific amino acid, allowing the ribosome to produce the wide variety of proteins that life requires.

This is the Central Dogma
The process of DNA being translated into RNA, and RNA being translated into proteins is the central dogma of biology. This process generates the wide diversity of life on Earth.



Materials and Reference Sheet

Each student will receive a bead cheat sheet and codon chart. Use the following color code:

DNA Bead Colors

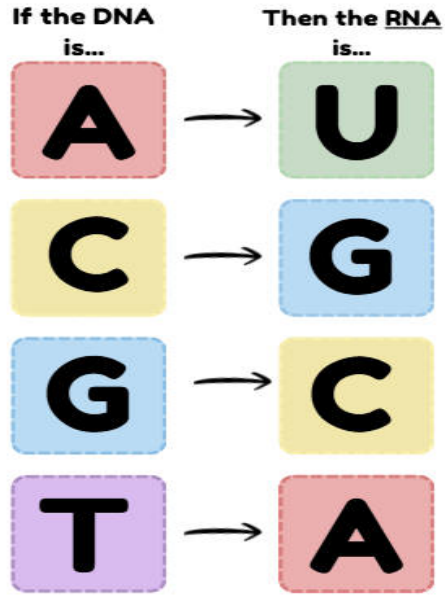
- A = Red
- T = Blue
- C = Yellow
- G = Green

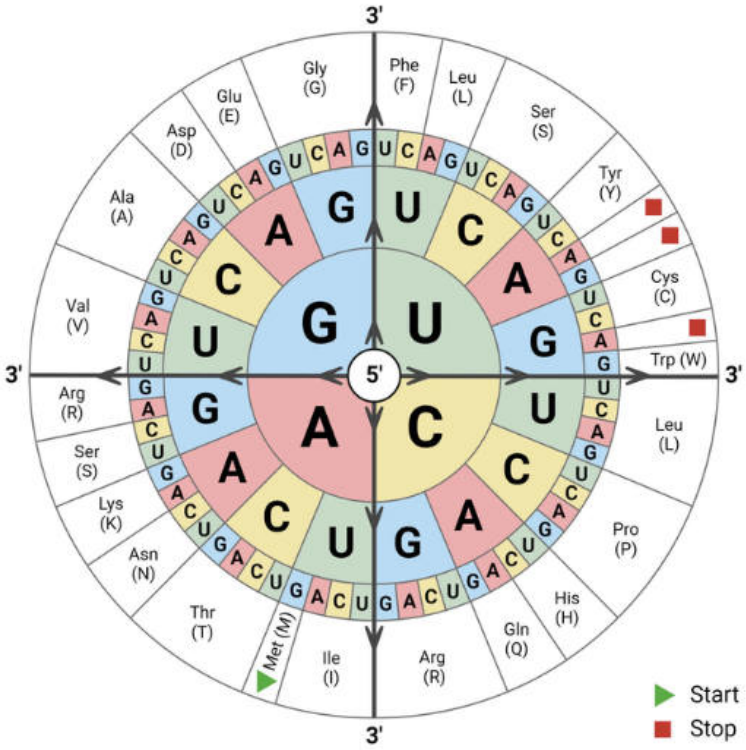
RNA Bead Colors

- U = Purple (Note: RNA uses U where DNA uses T)

Pairing Rules

During transcription, DNA is used as a template to build a complementary strand of RNA. Each DNA base pairs with a specific RNA base according to the following rules:





Codon to Amino Acid

A codon is a set of three RNA bases. Each codon matches to one specific amino acid. Use the chart to the right to find which amino acid your RNA codon codes for.

The Activity

After distributing materials and reviewing the reference sheet, students follow these steps to build their bracelet:

1. Pick 9 random DNA beads



2. Arrange them into 3 groups of 3 to create three DNA codons



3. Add a spacer bead to represent the transition from DNA to RNA



4. Using the transcription rules on your cheat sheet, match each DNA bead with its corresponding RNA bead



5. Use the codon chart to identify which three amino acids your RNA sequence codes for

6. Add a protein charm to the end and tie a knot to finish. Your bracelet is complete!



7. Scan the QR code on the next page to view a model of your protein and see how your amino acid sequence shapes its structure.

QR Code for Protein Structure:

Extension: Mutations Challenge

Small changes in DNA can change proteins and affect how cells function. Once students have built their original bracelet, invite them to try one of the following mutations if they are up for an additional challenge. Then, re-transcribe and re-translate to see what changed.

Can your mutation still make the same protein?

- Substitution (Swap): Change one DNA bead to another. Sometimes the protein stays the same (silent mutation) and other times it changes (missense mutation).
- Insert: Add a bead into your sequence. This shifts all following codons (frameshift mutation) and usually changes the protein.
- Delete: Remove a bead. Like insertion, it shifts codons and often changes the protein.

Try it: Apply a mutation, transcribe to RNA, translate to amino acids, and compare to your original protein.

Closing Discussion

To close, bring students together to reflect on what their bracelet represents. These are reminders on how the activity connects back to human biology.

- The order of DNA bases determines the amino acid sequence of a protein, just like the order of beads on your bracelet.
- Proteins perform essential roles: building tissues, transporting molecules, fighting infections, and enabling chemical reactions.
- Mutations in DNA can change proteins, sometimes leading to diseases such as sickle cell anemia or cystic fibrosis. Sometimes, these mutations have no effect at all as observed in the extension challenge.

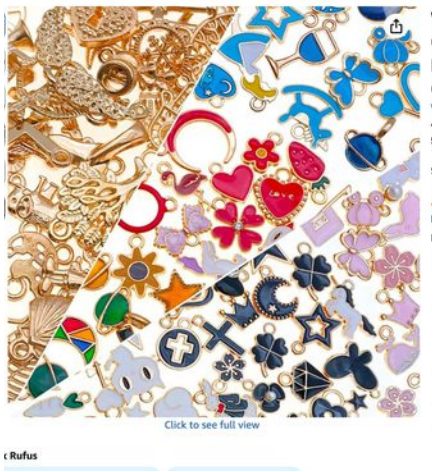
New link to explore popular proteins: https://alphafold.ebi.ac.uk/

- Note: I'm going to change the directions later because I changed the website

Links to materials:



https://tinyurl.com/2juhznak



https://tinyurl.com/b79yf744